

Spatial Interpolation Model (MKM-SP-001)

Wolfram Oracle Benchmark of the Code and its v2.0 Document

MKM Research Labs

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1 Executive Summary

This document benchmarks the Spatial Interpolation Model (MKM-SP-001): the module `src/models/floodrisk/spatial.py`. It asks two questions. Does the code compute what its model document (`spatial_model_v2.0.pdf`) says? And does what the document says hold?

Version 0.1 tested the filed v1.0 of that document. The code computed its distance and IDW exactly, but 10 of the document's 15 claims failed:

- The spatial index treated a degree of latitude and of longitude as the same length. It named the wrong nearest gauge in 176 of 1,000 queries over the Thames basin.
- The correlation was Euclidean degrees at 111 km a degree, not the equation printed.
- The coordinate path fixed the power and threshold.
- Nothing returned the nearest gauge.
- Nothing in the platform used the interpolation, the index or the correlation, and the tests, validation and kriging comparison the document cited did not exist.

The model was revised as v2.0:

- the index on the unit sphere, with a lookup returning the nearest gauge and its distance;
- the power and threshold taken from coordinates too;
- its constants moved to config;
- the document rewritten to say what the module computes, how accurate it is, and what uses it.

This version tests v2.0.

The code computes its documented equations. The haversine agrees with a 50-digit implementation of v2.0's equations to 9.0×10^{-10} m. IDW from distances and from coordinates, and the correlation, agree to 1.5×10^{-14}. The index names the nearest gauge in all 1000 queries (Table 1). Wolfram, computing each independently, agrees with the reference to 2.3×10^{-31}.

The document's claims hold: 13 of 13 (Table 2).

- All 27 figures in its tables reproduce from the code.
- Its statements about the model are true: both interpolation paths take the power, and the correlation's conversion overstates east–west separation by 61%. The sphere errs by the 0.32% it states, and nothing in the platform calls what it says nothing calls.
- Its statements about the evidence are true: the tests query the index, and there is no validation against observations and no kriging comparison.

Evidence. 8 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `ced02d123f9e`.
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-11-06 by `python -m tools.wolfram_oracle sp001`.

2 Model Purpose and Scope

2.1 What is benchmarked

Every function in `spatial.py`:

- `haversine_distance`;
- the interpolation, from coordinates (`spatial_interpolate_wse`) and from distances (`idw_interpolate_from_gauges`);
- the index (`build_spatial_index`, `nearest_gauge`);
- the correlation matrix (`build_correlation_matrix`);
- the projection onto a segment and a polyline (`nearest_point_on_segment`, `nearest_point_on_polyline`).

Its parameters are read from `config.models` (`_spatial.py`).

2.2 What is not

- Whether IDW or the correlation suits flood risk. v2.0 states that neither is validated or used (§5, Q5).
- The callers' own use of the distance and projection.

This is a benchmark of the implementation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

The reference implements v2.0's equations directly from its text:

- the haversine of Eq. 1, in its arctangent form, on a sphere of 6,371,000 m;
- the IDW of Eq. 2 with that distance and power p. The first gauge closer than the threshold (default 1 m) returns its own value;
- a gauge's water surface elevation as its depth plus its elevation (§3.2);
- the correlation of Eq. 3, on Euclidean degrees at 111 km a degree;
- the nearest gauge by Eq. 1 (§3.3);
- for the projection (§3.5), the true nearest point on a segment drawn straight in latitude and longitude, found by minimising the haversine distance along it.

4 Benchmark Method

The cases.

- IDW: six gauge sets at seven powers from 0.5 to 4, including gauges inside the threshold and on it. The coordinate path: four gauges around a target on the Thames, with the target on a gauge, 0.5 m from one, far off, and with a single gauge.
- The correlation: five properties within 15 km, $L = 2$ km, $\rho_0 = 0.6$.
- The index: the case v1.0 got wrong, a gauge 0.010° north and one 0.012° east of the target. Then 20 seeded random networks of 50 gauges over the Thames basin, queried at 50 random points each.
- The projection: four segments (east–west, north–south, diagonal and 1 km long). Points sit 10 m to 200 km from each midpoint, to the north, east and north-west. Nine points are projected onto

a river-like polyline.

The printed tables. Tables 2–5 were generated from the code. Each figure is judged as the code's value printed to the places the table shows.

The statements.

- What the model accepts and returns is judged from the code.
- What uses the model is judged by searching `src` and `scripts` for files that import the module and name each function.
- Q4, Q5 and Q7 are judged by searching the repository: for tests that query the index, for any code that calls the interpolation, and for any mention of kriging.

Wolfram. Wolfram works from each input's exact value, independently of the reference:

- the great-circle distance, as R times the angle between the two points' unit vectors (not the haversine);
- Eq. 2 at powers 2 and 5/2;
- Eq. 3 from exact degrees;
- the nearest point on the diagonal segment, where the squared chord along it is stationary (`FindRoot`);
- the true distance between six UK pairs under 100 km, by `GeoDistance` on WGS84.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Limit	Outcome
Eq. 1, the haversine: code against the reference, worst error (m)	9.0×10^{-10}	1.0×10^{-6}	pass
Eq. 2, IDW from distances, seven powers: worst relative error	1.8×10^{-16}	1.0×10^{-12}	pass
§3.2, IDW from coordinates: WSE as depth plus elevation, Eq. 1's distance: worst relative error	1.5×10^{-14}	1.0×10^{-12}	pass
Eq. 3, the correlation at 111 km a degree: worst relative error	1.3×10^{-15}	1.0×10^{-12}	pass
§3.3, the index over 1000 queries: nearest gauges that differ from Eq. 1's	0	0	pass
§3.3, the index's distance: worst relative error	7.6×10^{-13}	1.0×10^{-9}	pass
§3.5, the polyline projection picks the reference's segment (9 points): segments that differ	0	0	pass
Wolfram agrees with the reference	2.3×10^{-31}	1.0×10^{-25}	pass

5.1 The model document's claims

Table 2: The model document's claims, tested

Claim	Stated	Benchmark finds	Outcome
Table 2	WSE by IDW power	7 of 7 figures reproduced	met
Table 3	WSE by distance of the first gauge	7 of 7 figures reproduced	met
Table 4	WSE by number of gauges	5 of 5 figures reproduced	met
Table 5	the projection's worst error by distance from the line	8 of 8 figures reproduced	met
§3.2	both interpolation paths take the power and the threshold	both do	met
§3.2	with no gauges, the interpolation returns 0	both paths do	met
§3.3, §4.2	the index returns the nearest gauge by Eq. 1, and its distance	1000 of 1000 queries; distance to 7.6×10^{-13}; the gauge 831 m away over one 1112 m away	met
§3.4	the correlation converts degrees at 111 km, overstating east–west separation by 61% at 51.5°N	exact to 1.3×10^{-15}; 60.6%	met
§5	the sphere errs by up to 0.32% against WGS84 for UK pairs under 100 km	worst -0.32%	met
§2.2, Table 1	nothing calls the interpolation, the index or the correlation; the haversine and projection serve the gauge, property and seismic code	so: gauge, property, seismic	met
Q4	the unit tests query the index	they do	met
Q5	nothing validates the interpolation against observations	no code calls it	met
Q7	nothing compares it with kriging	kriging appears nowhere in the repository	met

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The code is shown to compute its documented equations, and the document to describe it. Whether IDW represents a flood surface is not tested. v2.0 says so, and nothing in the repository tests it.

2. **Searches are of this repository.** "Nothing calls it" means nothing in PhysicalRisk at the commit above.

3. **The index is judged on random networks** over the Thames basin. It is exact by construction (§3.3), so the networks test the code, not the method.

7 Document History

Date	Version	Description	Author
06-Nov-2026	0.1	First issue: tests the filed v1.0; 10 of its 15 claims fail (not filed)	David K Kelly
06-Nov-2026	0.2	Tests v2.0, the revised model	David K Kelly